ENVIRONMENTAL CHEMISTRY

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Environmental Chemistry

Environmental chemistry is the study of chemical phenomena occurring in the environment.

Introduction

The environment means surroundings. It has four following components.

I. Atmosphere:

(A) Role of atmosphere in life:

(i) All essential gases present in atmosphere must for life on earth.
(ii) Water molecules carried in the medium of atmosphere.
(iii) Harmful ultra violet radiations absorbed in the medium of ozone layers, saving life of mankind.
(iv) Saving life by absorbing undesired, excess heat coming from sun. Also key security layer from global warming conditions.

(B) Important data related to atmosphere (Remember All):

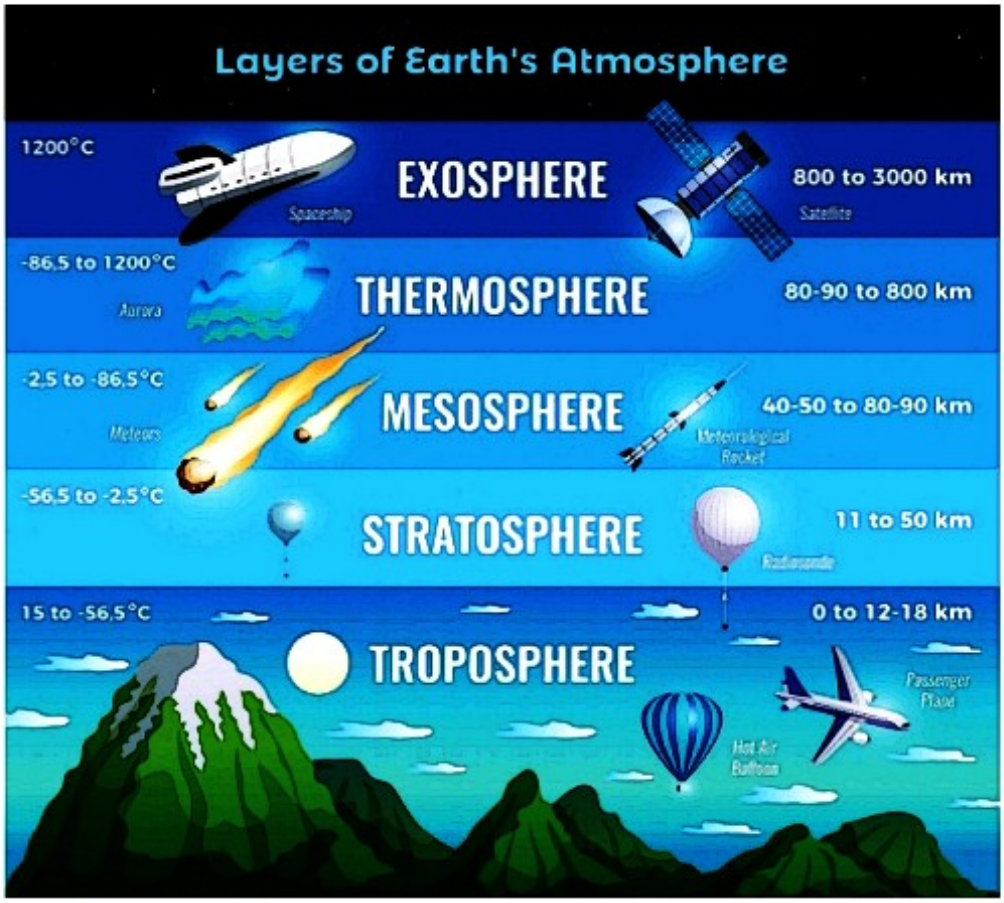
(i)	Pressure of atmosphere	10^5 N/m^2
(ii)	Mass of atmosphere	5×10^{15} tonnes.
(iii)	Temperature	– 100°C to 1200°C

Note :
The pressure and the density of air decrease by one half with 5 km increase in altitude.

(C) Composition of air (or atmosphere):

It is divided in 3 categories :

(i) Major components = N_2, O_2, water vapour.
(ii) Minor components = Ar, CO_2
(iii) Negligible constituent = He, Ne, Kr, CH_4, H_2, CO, N_2O, SO_2, NO, NO_2, HCHO, NH_3, O_3.



D. Layers of atmosphere:

It has 4 particular layers. These regions are defined by the temp.

REGION	ALTITUDE FROM EARTH'S SURFACE	TEMPERATURE RANGE	SPECIES PRESENT OR GASES PRESENT
Troposphere	0 - 11 km	decrease from 15 to (– 56°C)	N_2, O_2, CO_2, water vapour
Stratosphere	11 - 50 km	increase from (– 56°C) to (– 2°C)	N_2 , O_2 O_3 , 0-atm
Mesosphere	50 – 85 km (ionosphere)	decrease from (–2°C) to (-92°C)	N_2, O_2, NO^+, O_2^+
Thermosphere	85 – 500 km (ionosphere)	increase from (–92°C) to (1200°C)	O_2^+, O^+, NO^+

II. Hydrosphere :

75% part covered with hydrosphere inform of ocean , river, lake and other water resourses .

III. Lithosphere :

Most common part on earth suitable for living on. We built house, roads, school everything on lithosphere . Hard rock, field, plane ground or velly everything is part of lithosphere.

IV. Biosphere

Biological space including atmosphere, hydrosphere and lithosphere where all living organism intract and live together , called as biosphere.

Environmental pollution & environmental pollutant

Environmental pollution is defined as direct or indirect undesirable changes in our surroundings because of human activities or natural happenings. It has harmful effects on animals, plants, and humans.

Pollutants can be defined as pollution causing substance. It may be solid, liquid, or in gaseous form which are present in greater concentrations than their expected amount in nature. These are produced due to natural happenings or due to human activities.

According to nature of pollutants -

(1) Fast Degradable: Which rapidly break down by natural processes. For example, discarded vegetable.

(2) Slowly Degradable: Waste management is serious issue , everyone taking about plastic waste , nuclear waste , heavy metals. All these remain for decades in environment For example, dichlorodiphenyltrichloroethane (DDT). We haven't any process to treat them, existance in atmosphere is serious issue for living organism.



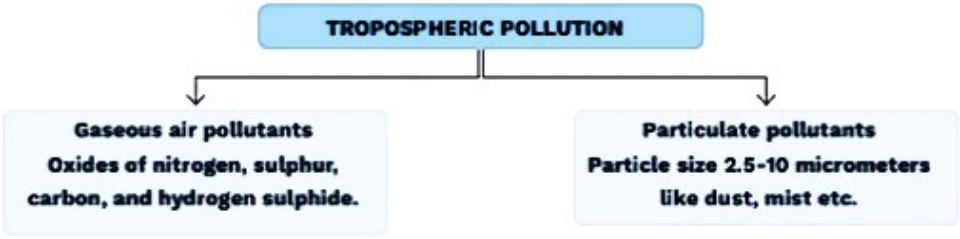
The procedure of environmental contamination comprises:

(i) The origination of pollution causing substances from diverse sources

(ii) Transportation by air or water or dumping into soil by human being.

Tropospheric pollution:
It is observed due to the availability of unwanted particles in air in their solid or gaseous form.





Gaseous air pollutants:

(a) Oxides of sulphur:

$$\text{Sulphur (from fossil fuel)} \xrightarrow{\text{Burn}} \text{Oxides of sulphur } (SO_2, SO_3)$$

Toxic effect of oxides of sulphur:



Considerable and serious effect on health of living organism. It can cause asthma, bronchitis, emphysema like respiratory ailments in human being even when present in very low concentration.

(i) It causes irritation to human eyes giving tears and redness.

(ii) Its higher concentration is held responsible for the stiffness of flower buds. It finally eventually their fall off from plants.

$$2SO_2(g) + O_2 \xrightarrow[\text{Matter (catalyst)}]{\text{Particulate}} 2SO_3$$

$$SO_2(g) + O_3(g) \longrightarrow SO_3(g) + O_2(g)$$

$$SO_2(g) + H_2O_2(\ell) \longrightarrow H_2SO_4(aq)$$

The SO_2 present in the atmosphere is generally contaminated with SO_3. The trioxide so formed gets converted into sulphuric acid in the presence of moisture. The sulphuric acid in the atmosphere forms acid rain.



(b) Oxides of nitrogen:

At normal temperature, N_2 and O_2 can not combine.But they can react at high altitudes when lightning strikes, to form oxides of nitrogen. NO_2 is even oxidised to nitrate ion which is washed into soil, and finally serving as fertilizer. At the time of combustion of hydrocarbons in automobiles, some N_2 and O_2 combine to give significant amount of NO and NO_2.

$$N_2(g) + O_2(g) \xrightarrow{1483K} 2NO(g)$$

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)\,(\text{slow reaction})$$

$$NO(g) + O_3(g) \longrightarrow NO_2(g) + O_2(g)\,(\text{faster reaction})$$

Toxic effect:

(i) Due to oxides of nitrogen, red haze is observed in the traffic and congested places.

(ii) Higher concentration of nitrogen dioxide harms the leaves of plants and thus decreases the rate of photosynthesis.

(iii) It is harmful for living being.

(iv) NO_2 is also harmful to various textile fibres and metals.

(c) Hydrocarbons:

Perotl, diesel, kerosene etc. fossil fuels combustion in Automobile, Agriculture, Factory productions are key source of hydrocarbon pollutants.

Toxic effect:

(i) It is responsible for Cancer, a very deadly disease.

(ii) Stoping growth of plants, breaking of tissues, ageing.

(d) Oxides of carbon:

(1) Carbon monoxide: Sources of carbon monoxide:

(i) Incomplete combustion of carbon, coal, firewood, petrol etc.

(ii) Auto mobile exhaust.

Toxic effect:



(i) Carbon monoxide is toxic. It binds to haemoglobin in the blood to forma complex, carboxy haemoglobin. The complex is about 300 times more stable than the oxy-haemoglobin. This stable complex reduces the oxygen carrying capacity of blood to greater extent. This causes oxygen deficiency which leads to headache, weak eyesight, unconsciousness and even death.

(ii) For pregnant women increased CO level can cause different ailments like premature birth, spontaneous abortions and deformed babies.

Note :

Sink of CO: A large amount of CO are added in the atmosphere.Micro organisms present in the soil convert CO into CO_2 so the level of CO does not rise too much. The microorganism (bacteria) acts as sink for CO.

(2) Carbon dioxide [CO_2]: Sources of carbon dioxide:

(i) Respiration) (exhaled air contains CO_2).
(ii) Combustion of fossil fuels for energy.
(iii) Decomposition of limestone during the manufacture of cement.
(iv) Volcanic eruptions.

It is found in very small amount(0.03 percent) by volume in the atmosphere. It is confined to troposphere only.It is essential for photosynthesis and thus provides oxygen.This process helps in maintaining the delicate balance. But other factors (like deforestation and burning of fossil fuel) also increase the CO_2 level and disturb the atmospheric balance. This increased amount is held responsible for global warming.



Global Warming and Green-house effect

It is estimated that 25% of the solar energy coming to the earth radiates back to the atmosphere. The remaining part is trapped by the earth surface. Some fraction is also absorbed by gases present in the atmosphere. The main absorbent gases are methane, CO_2, CFCs, ozone and water vapour. Thus, they increase the temperature of the atmosphere which finally leads to global warming.

It is always said that we humans are living in a greenhouse due to the thick layer of atmospheric air surrounding us which provides constant temperature since long time. (However, it is undergoing slow change).



In this manner, our atmosphere is working as natural greenhouse maintaining the temperature and thus making our earth perfect for life.
Greenhouse gases include methane, CO_2, N_2O, CFCs, ozone and water vapour. Among all these gases CO_2 is the main contributor.

Greenhouse gases	Source
CH_4	Paddy fields (due to burning of vegetations) ,Coal mines, fossil fuel
CFCs	Man-made industrial chemical mainly for aerosol propellant refrigeration and air- conditioning.
N_2O	Environmental gas

Due to uncontrolled use of fertilizers and fossil fuel burning, the amount of greenhouse gases have increased significantly.

Increase in average global temperature can have disastrous environmental (melting of Glaciers and polar ice caps) and human health hazards (malaria, dengue, fever and other infectious diseases)

Acid Rain:

When rain water reacts with atmospheric CO_2 in the presence of H^+ ions , the following reactions take place:

$$H_2O(\ell) + CO_2(g) \rightleftharpoons H_2CO_3(aq):$$
$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

When the natural rain falls through polluted air, it also comes in contact with gaseous oxides of sulphur (SO_x), and nitrogen (NO_2), mist of hydrochloric acid and phosphoric acid..

These substances dissolve in falling rain making it more acidic. When the pH of rainwater drops below 5.6, it is called acid rain.
For rain water, pH = 5.6
For acid rain, pH < 5.6



Nowadays, the term acid rain is also used to describe different ways of deposition of acid on earth surface. Due to movement with blowing wind, the oxides of sulphur and nitrogen can finally settle (in dry form)on the earth surface or can be found as wet deposition in snow, water and fog.

Special Note : pH of normal rain water = 5.6(due to dissolved CO_2 in water).

Chemistry of acid rain: Nitrogen and sulphur oxides are major contributors in acid rain. Oxidation of NO_2 and SO_2 produces acid (H_2SO_4 and HNO_2). Major sources for sulphur or Nitrogen oxides are following.

(i) **Burning of fossil fuels** (which contain sulphur and nitrogenous matter) such as petrol and diesel in motor engines or oil and coal in power stations and furnaces.

(ii) **Volcanic eruptions, forest fires and bacterial decomposition of organic matter**

$$2SO_2(g) + O_2(g) + 2H_2O(\ell) \rightarrow 2H_2SO_4(aq.)$$

$$4NO_2(g) + O_2(g) + 2H_2O(\ell) \rightarrow 4HNO_3(aq.)$$

Ammonium salts also produced, observed in the form of atmospheric haze (Aerosol of fine particles).

Harmful effect of acid rain:

(i) Soil losses all important nutrient dissolved in, wash away with acid rain.

(ii) Rainwater with low pH valve (acidic water) not suitable for aquatic life, fishes die in acid rainwater, balance in ecosystem get disturbed.

(iii) Respiratory illness effect in long term exposer in acid rains.

(iv) Corrosion in iron pipes, make unfit water for drinking purpose. Heavy metals dissolve in water.

(v) Acid rainwater react with marble, dolomite and metals. It will hamper life spam of buildings, soluble compounds formed wash away with rain.

$$[CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 \uparrow + H_2O]$$

Particulate pollutants:

It refers to very small solid or liquid particles suspended in air. These are found in dust particles, smoke particles from fires, ash from industries and vehicle emissions.



Particulates present in the atmosphere may be viable or non-viable.

Viable Particulates: Allergic reactions to human beings is due to minute living organism dispersed in the atmosphere. Some examples are fungi, bacteria , fungi , moulds and algae.

Non-viable Particulates: Different form based on nature and size:

Smoke particulates:	Burning of organic matter gives off solid and liquid particles.
Dust:	Crushing and grinding of solid materials which gives minute particles (particle size over 1 micrometer in diameter). Example Blusting in rock produces dust particles.
Mist:	By Vapour condensation and particle produces by spray . example Mist , Pesticides herbicides etc.
Fumes:	These are usually formed by condensation of vapours. Example Acid bottle opening in lab shows white fumes on top of bottle.

Toxic effect:

(i) 5 microns or bigger particle pollutants are supposed to lodge in the nosal passage.

(ii) 1.0 micron particle pollutant easily enter in lungs .

(iii) Leaded petrol is the primary source of air-borne lead emission. Lead is the heavy metal which interferes with the development and maturation of heamatids(RBC's).

Smog

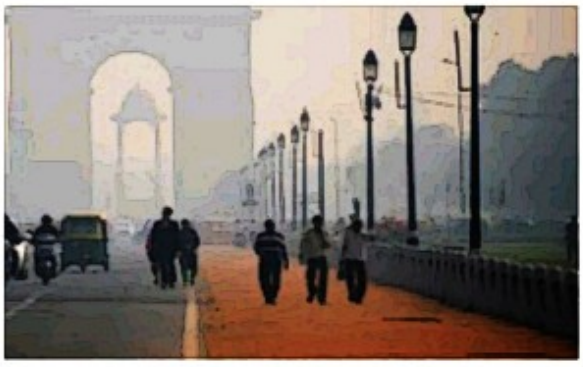
Smog = Smoke + Fog

The name suggests that it is combination of smoke and fog. It is one of the most common examples of air pollution.

It is of two types:

Classical Smog:

Its components are smoke, fog and sulphur dioxide. It is known as London smog (At first it was observed in London) or sulphurous smog. Due to its reducing nature(because of the presence of SO_2 and carbon /soot particles) in chemical reactions, it is also known as reducing smog. It occurs in cool and humid climate.



Photochemical Smog:

It is also known as Los Angelas smog (as first occurred in Los Angelas). It occurs in dry, warm and sunny climate. The main components of this smog result from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Photo chemical smog has high concentration of oxidising agent like NO_2 and is, therefore called as oxidising smog.

Mechanism of photochemical smog:

* Burning of fossil fuels give many pollutants. Most important pollutants are hydrocarbons (unburned fuels) and nitric oxide (NO).
* When the concentration of these pollutants increases to sufficiently high levels, a chain reaction takes place .In this reaction, nitric oxide is converted into nitrogen dioxide. This nitrogen dioxide decomposes in the presence of sunlight forming NO and free oxygen atom.

$$NO_2(g) \xrightarrow{h\nu} NO(g) + O(g) \quad \text{(i)}$$

Due to high reactivity, oxygen atoms combine with atmospheric oxygen forming ozone.

$$O(g) + O_2(g) \rightleftharpoons O_3(g) \quad \text{(ii)}$$

This O_3 obtained reacts with NO(g) formed in the reaction (i) to regenerate NO_2 . NO_2 is a reddish brown/brown gas .At sufficiently high level , it forms haze.

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g) \quad \text{(iii)}$$

* Like NO_2, O_3 is a toxic gas. Both NO_2 and O_3 are strong oxidising agents. With unburnt hydrocarbons in the polluted air ,they form compounds like formaldehyde, acrolein and peroxyacetyl nitrate (PAN).

$3CH_4 + 2O_3 \longrightarrow 3CH_2{=}O + 3H_2O$

$CH_2{=}CHCH{=}O$ (acrolein)

CH_3COONO_2 (peroxyacetyl nitrate,PAN)

$$O_3 \rightarrow O_2 + [O]$$

$$O + \text{hydrocarbons} \longrightarrow RC\dot{O} \text{ (free radical)} \xrightarrow{O_2} RC\dot{O}_3 \text{ (peroxyl acyl free radical)}$$

$$RC\dot{O}_3 + \text{hydrocarbons} \rightarrow CH_2 = O$$

CH_2 = CHCHO, ketones etc

$$RC\dot{O}_3 + O_2 \rightarrow RC\dot{O}_2 + O_3$$

$$RC\dot{O}_3 + NO \rightarrow RC\dot{O}_2 + NO_2$$

$$[R - \overset{O}{\overset{\|}{C}} - O - \dot{O} + NO_2 \rightarrow R - \overset{O}{\overset{\|}{C}} - O - O - NO_2 \text{ (PAN)}]$$

Harmful/toxic effects of photochemical smog:

(i) NO and O_3 causes various health problems for human being.They can cause irritation to the throat and nose .At higher concentration,it gives headache, coughing, and breathing problem.

(ii) ozone and PAN also causes irritation to our eyes.

(iii) It leads to severe damage to plant life. PAN inhibits photo system, inhibits formation of chlorophyll and also spoils enzymes.

(iv) It also causes corrosion of metals, stones, building materials, rubber and painted surfaces.

Control of photochemical smog:

(i) By controlling its primary factors like NO_2 and hydrocarbons and secondary factors like O_3 and PAN .

(ii) Using catalytic converters in the automobiles.

(iii) By plantation of certain plants like, pinus, juniparus, quercus, pyrus ant vitis which metabolise NO.

Stratospheric Pollution:

Production and Decomposition of Ozone:

Ozone (O_3) is found in upper stratosphere with significant amount. The ozone layer shields us from potentially harmful ultra violet (UV) rays coming from the sun.

These UV radiations can cause skin cancer (known as melanoma) and therefore, we need to maintain ozone layer.

Ozone is produced as result of the action of ultraviolet radiations on dioxygen as given below:

$$O_2(g) \xrightarrow{UV} O(g) + O(g)$$

$$O(g) + O_2(g) \underset{\text{dynamic equilibrium}}{\overset{UV}{\rightleftharpoons}} O_3(g) \text{ (thermodynamically unstable)}$$

Freons are held responsible for depletion of ozone layer. Freons are chlorofluorocarbon compounds (CFCs).

They are found to be nonreactive, non-flammable, non toxic carbon compounds. They are known for their use in refrigerators and air conditioners.They can be used in the production of plastic foam as well as for cleaning computer parts.

After their release in atmosphere, these CFCs intermix with other atmospheric gases and reach to stratosphere where they cause depletion in ozone layer as described below:

$$CF_2Cl_2(g) \xrightarrow{UV} \dot{C}l(g) + \dot{C}F_2Cl(g)$$

$$\dot{C}l(g) + O_3(g) \longrightarrow Cl\dot{O}(g) + O_2(g)$$

$$Cl\dot{O}(g) + O(g) \longrightarrow \dot{C}l(g) + O_2(g)$$

In this the chlorine radicals are regenerated continuously and cause the decomposition of ozone.

Hence, Freons act as agents for generation of chlorine radicals regularly . In this way, it is damaging the ozone layer of stratosphere.

Ozone Layer Depletion:

Scientists observed ozone layer depletion over south pole in 1980s. Some set of conditions were identified for it.

Polar Stratospheric Clouds:

(i) In summer, NO_2 reacts with chlorine monoxide and CH_4 reacts with chlorine atoms forming chlorine sinks . It controls ozone depletion.

(ii) In winter,polar stratospheric clouds are observed in Antarctica. These are special type of clouds.



These clouds provide surface for hydrolysis of Chlorine nitrate forming HOCl (Hypochlorous acid) . Clorine nitrate gives Cl_2 on reaction with HCl.

$$\dot{C}lO(g) + NO_2(g) \rightarrow ClONO_2(g);$$

$$\dot{C}l(g) + CH_4(g) \rightarrow \dot{C}H_3(g) + HCl(g)$$

$$ClONO_2(g) + H_2O(g) \rightarrow HOCl(g) + HNO_3(g)$$

$$ClONO_2(g) + HCl(g) \rightarrow Cl_2(g) + HNO_3(g)$$

(iii) In spring, when sunlight returns to the Antarctica, it breaks up the clouds due to its heat. Photolysis of HOCl and Cl2 occurs in the presence of sunlight according to the following reactions.

$$HOCl(g) \xrightarrow{h\nu} \dot{O}H(g) + \dot{C}l(g); Cl_2(g) \xrightarrow{h\nu} 2\dot{C}l(g)$$

These chlorine radicals are responsible for initiating the chain reaction for depletion of ozone layer.

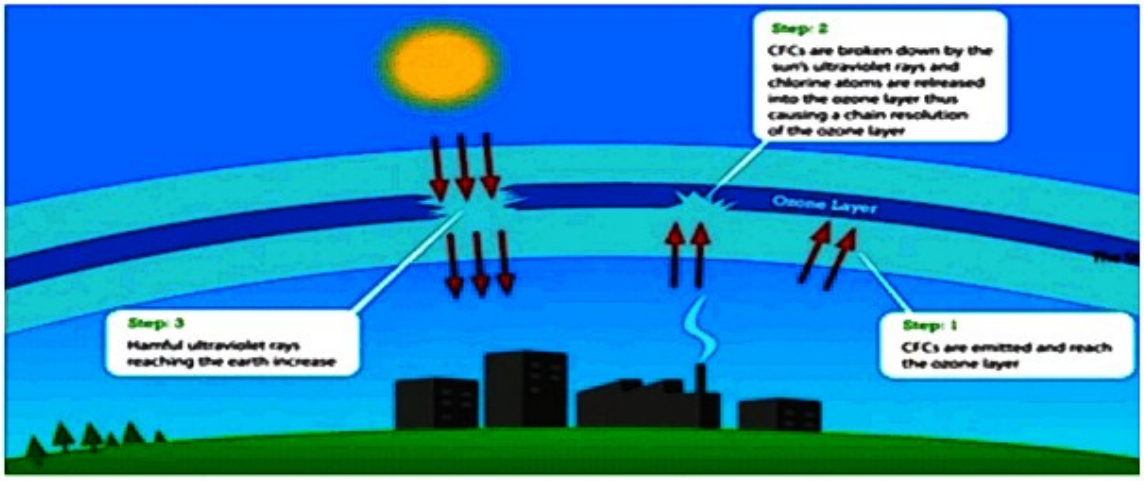
Harmful effects of depletion of ozone layer:

(i) UV radiations can cause various problems like ageing of skin, cataract, sunburn, skin cancer, killing of many phytoplanktons, damage to fish productivity etc. .

(ii) UV radiations affects the plant protein which eventually leads to the harmful mutation of cells.

(iii) UV radiations increases the rate of evaporation of water through the stomata of the leaves. It decreases the moisture of the soil.

(iv) Paints and fibres lose their shine due to UV radiations.



Example:

Q1 **Explain tropospheric pollution in 100 words.**

Ans Tropospheric pollution is caused by the existence of undesirable gases and the solid particles in the air. The major gaseous and the particulate pollutants present in the troposphere are as follows:

- Gaseous air pollutants: These include mainly oxides of sulphur (SO_2, SO_3), oxides of nitrogen (NO, NO_2) and oxides of carbon (CO,CO_2) in addition to hydrogen sulphide (H_2S), hydrocarbons, ozone and other oxidants.
- Particulate pollutants: These include dust, mist, fumes, smoke, smog, etc.

Q2 **What is the composition of photochemical smog? Why is it so called?**

Ans Mixture of NO_2, O_3, peroxyacyl nitrates (PAN), aldehydes, ketones, hydrocarbons, and CO.

Water Pollution:

Water pollution mainly refers to contamination of water due to human activities. .



(i) Point source pollution:
In this case, we can identify the source or place of pollution very easily. Examples are municipal and industrial discharge pipes where pollutants are added to the water source.

(ii) Non point source pollution:
In this case, we can not identify the source of pollution very easily. Examples are agricultural waste from farmland and animals), acid-rain,water-drainage (coming from streets, parking area and lawns) etc.

Water pollutants and their sources.

	Pollutant	Source
(i)	Micro-organisms	Domestic sewage
(ii)	Organic wastes	Decaying animals and plants, domestic sewage, animal excreta and waste, discharge from food processing factories
(iii)	Plant nutrients	Chemical fertilizers.
(iv)	Toxic heavy metals	Industries and chemical factories.
(v)	Sediments	Erosion of soil by strip mining and agriculture.
(vi)	Pesticides	Chemical used for killing insects, fungi and weeds.
(vii)	Radioactive substances	Mining of uranium containing minerals.
(viii)	Heat	Water used for cooling in industries.

Origin of water pollution:

(a) **Pathogens (Infection or disease causing agents) :** Pathogens are disease causing agents. It includes bacteria and other organisms. They are found in domestic sewage and human excreta. They are considered as most harmful water pollutants. Bacteria (such as Streptococcus faecalis and Escherichia coli) present in human excreta cause gastrointestinal diseases.

(b) **Organic waste:** Grass, leaves, trash and other organic matter are also considered as strong water pollutants. After decay,they pollute water. All these are biodegradable. Excessive phytoplankton growth within the water also causes water pollution.

Biodegradable waste: These are domestic wastes. It can be rapidly decomposed by micro organisms.

Non-Biodegradable waste: These are poison wastes that do not degrade or degrade very slowly in the ecosystem naturally. They are not recycled in the natural environment.

Dissolved Oxygen (DO):

Large number of bacteria is found with organic matter in water.They decompose organic matter.Here they also consume dissolved oxygen.But the amount of oxygen (DO) is limited in water. It can go up to 10 ppm in cold water. In atmospheric air, its concentration is 200,000 ppm.So, decomposition of organic matter can cause depletion of DO in water. DO is also crucial for aquatic life . For growth of fish, DO concentration should be above 6 ppm.

DO comes from atmosphere or from photosynthesis process.At night,the amount of DO gets reduced because at that time photosynthesis stops but respiration of plants continues.

Decomposition of organic matter requires dissolved oxygen. In case of excess organic matter, all the available oxygen will be consumed. This leads to death of oxygen-dependent aquatic life.



We can understand the importance of DO for aquatic life. In absence of DO, anaerobic bacteria (which do not need oxygen) start the decomposition of organic matter/waste. It produces chemicals with foul smell . These are harmful to human health. Aerobic oxygen requiring bacteria degrade these organic waste and keep the water depleted in dissolved oxygen

Biochemical Oxygen Demand (BOD):

It represents oxygen requirement by bacteria for decomposition of organic waste present in a given volume of water .

Higher value of BOD means highly polluted water. For Clean water, BOD value is less than 5 ppm whereas BOD value of 17 ppm or more shows highly polluted water.

(c) Chemical pollutants:

It include cadmium, mercury, nickel, etc.These are water soluble heavy metals. All these are toxic to humans. It can not be excreted out. Over time, increase in their concentration beyond tolerance limit, they can cause harm to kidneys, central nervous system, liver, etc.

Some acids and salts are found in drainage coming from mines.For example: Sulphuric acid, sodium salts , calcium salts etc. They can be used for melting snow particles in winters.These are water soluble .These are regarded as chemical pollutants.

Organic pollutants: Oil spilling in water-bodies is a major concern. Soap and detergents used in household are also polluting .Fertilizers and Pesticides used for agricultural purpose add organic pollutants. Fertilizers can support the growth of microorganisms which can deplete DO in water.IT makes problem for aquatic life like increase in decay rate and death.In industries, poly chlorinated biphenyls used as cleansing solvent. It can cause cancer.

Eutrophication: Eutrophication: This process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as Eutrophication.

Limit of chemicals in drinking water:

(a) Fluoride:

Its concentration can be up to 1 ppm or 1 mg dm^{-3}.It is essential for healthy teeth. Its deficiency as well as high concentration is unhealthy.Its deficiency causes tooth decay.. Since the outer layer of teeth contains hydroxyapatite $[3Ca_3(PO_4)_2 .Ca(OH)]$, it will be changed to fluorapatite $[3\ Ca_3(PO_4)2.CaF_2]$ by fluoride ions. It makes tooth enamels stronger But if concentration goes beyond 2 ppm, it gives brown mottling to our teeth. Similarilarly its concentration above 10 ppm makes our bones and teeth weaker.

(b) Lead:

The upper standard limit of lead concentration is about 50 ppb. It can damage kidney, liver, reproductive system etc.

(c) Sulphate:

With low concentration, it makes no problem.Beyond 500 ppm ,it makes stomach problem.

(d) Nitrate:

Its limit is upto 50 ppm .Higher concentration gives **methemoglobinemia ('blue baby' syndrome).**

(e) Other elements:

Elements	Upper limit (ppm or mg dm^{-3})
Iron	0.2
Manganese	0.05
Aluminium	0.2
Copper	3.0
Zinc	5.0
Cadmium	0.005

Soil Pollution:
Insecticides, pesticides and herbicides cause soil pollution.

(a) Pesticides:
These are chemicals used for pest control. These ars harmful and affect our ecosystem badly. If used regularly, pests become resistant to that set of pesticides. Thus these pesticides become inefficient. When pests developed resistance to DDT, aldrin and dieldrin came into use.



These are insoluble in water and non-biodegradable. These are transferred from lower level to higher level via food chain. If their concentration increases beyond certain level, they make serious metabolic and physiological problems.

(b) Herbicides:
Herbicides such as sodium chlorate ($NaClO_3$), sodium arsenate (Na_3AsO_3) and many other are now a days used to kill weeds. However because of their toxic effect to mammals, their use have been restricted. Some herbicides cause birth effect. Organic compounds such as triazines are widely used as herbicides in these days.

(c) Insecticides:
These are used to control insects and thus help to curb diseases and protect crops. They include chlorinated hydrocarbons, DDT, BHC and malathion etc. However, due to adverse effect, use of DDT has been banned in India.

Industrial waste:

(i) Biodegradable waste: Generated by paper mills, cotton mills, food processing units and textile factories.

(ii) Non-Biodegradable waste: Generated by integrated iron and steel plants (blast furnace slag and steel melting slag); thermal power plants (fly ash); aluminium, zinc and copper manufacturing (mud and tailings); Fertilizer industries (gypsum)

Disposal: In cement manufacturing, fly ash obtained from power plants and slag from the steel industries are used.

Using incineration, large deposits of harmful waste can be controlled. Industrial garbage is burnt.

Methods to control environmental pollution:

The management of waste:

The production and improper discharge of waste are responsible for environmental pollution. Besides the household waste, which includes sewage and municipal trash, numerous harmful industrial wastes from manufacturing processes need treatment and proper disposal.

Recycling:

It is usually said that recycling changes waste into wealth. It provides cost saving on crude materials as well as waste disposal.

For example: Recycling of glass, utilization of scrap metal, generating energy from combustible waste etc.

Sewage Treatment:

The main steps are mentioned below:

1. The removal of large solids by filtering the wastewater through screens. The solids that are removed are discarded in landfill sites.
2. Removal of solids that settle out (called sludge). This process can be used for removal of grease, which can be skimmed off.
3. Decomposition of organic waste by micro organisms.
4. Using various physical and chemical processes, the quality of wastewater can be improved.



Incineration:

Incineration converts organic materials to CO_2 and H_2O. The main disadvantage of incineration is that it leads to air pollution. Further, the ash from municipal incinerators is very finely divided and can be ingested into the lungs. Incomplete combustion of PCBs can cause the formation of highly toxic chloro compounds such as polychloro dibenzodioxin (PCDDs) and polychloro dibenzofurans (PCDFs).

Digestion:

It refers to decomposition of wastes by micro organisms in the absence of oxygen. It may be used in the treatment of sewage sludge and also for decomposing a variety of toxic organic wastes. The complete process is the conversion of the organic material into carbon dioxide and methane i.e.

$$2(CH_2) \rightarrow CO_2(g) + CH_4(g)$$

These products can be used as fuel.

Dumping:

Usually the sludge contains phosphorus and nitrogen which makes it useful as a fertilizer. But sometimes they contain highly toxic metals also so their dumping should be done carefully.



Collection and disposal:





Green Chemistry:

(i) It refers to all the methods and principles which reduces environmental pollution.
(ii) It gives mechanism which brings minimum pollution to our environment.
(iii) In chemical reactions, conditions should be optimised such that pollutants could not be generated. It will be environment friendly and cost-effective.

Green Chemistry in day-to-day life:

(i) Dry cleaning of clothes: For dry cleaning, liquified CO_2 is used in place of tetra chloroethene. Tetra chloroethene pollutes ground water. It is also a suspected carcinogen. Liquid CO_2 makes less harm. Now a days H_2O_2 is also used. It takes less amount of water.

(ii) Bleaching of paper: Use of H_2O_2 with suitable catalyst has replaced the use of chlorine gas for bleaching of paper.

(iii) Synthesis of chemicals:

$$CH_2 = CH_2 + O_2 \xrightarrow[Pd(II)/Cu(II)(in\ water)]{Catalyst} CH_3CHO$$ (90%, one step method)

Important: Green chemistry leads to reduction in:

(i) material.
(ii) energy consumption and
(iii) waste generation.

Example

Q1 **Which of the following statements is true?**

(A) London smog is oxidising in nature.
(B) London smog contains H_2SO_4 droplets.
(C) London smog is mixture of smoke, fog and SO_2.
(D) London smog causes bronchitis.

Ans (C)
London smog contains smoke, fog, and SO_2.

Q2 **Which of the following processes doesn't increase the amount of CO_2 in atmosphere?**

(A) Decay of animals **(B) Breathing**
(C) Photosynthesis **(D) Burning of petrol**

Ans (C)
Photosynthesis does not increase the amount of CO_2 in atmosphere.

Q3 **Consider the following statements and select the correct option:**

S_1: Dust is non-viable particle.

S_2: Particulates acquire negative charge and are attracted by the positive electrode.

S_3: O_2 is a green house gas.

S_4: Algae is a viable particulate.

(A) S_1 and S_2 only **(B) S_1, S_2 and S_3 only**

(C) S_1, S_2 and S_4 only **(D) S_2, S_3 and S_4 only**

Ans (C)

Viable particulates are small size living organisms like fungi, bacteria, moulds, algae etc. Non-viable particulates are formed by disintegration of large size materials or condensation of small size particles or droplets e.g., mist, smoke, fumes and dust.

Q4 **Among the following statements, which one is true about ozone layer?**

(A) It is harmful because ozone is dangerous to living organism.

(B) It is beneficial because oxidation reaction can proceed faster in the presence of ozone.

(C) It is beneficial because ozone cuts off the ultra violet radiation of the sun.

(D) It is harmful because ozone cuts out the important radiation of the sun which are vital for photosynthesis.

Ans C

It is beneficial because ozone cuts off the ultraviolet radiation of the sun.

Q5 **Incomplete burning of petrol or diesel oil in automobile engines can be best detected by testing the fuel gases for the presence of :**

(A) CO and water vapour **(B) CO**

(C) NO_2 **(D) SO_2**

Ans B

Incomplete combustion of petrol or diesel oil produces CO gas.

Q6 **Which of the following acts as a sink for chlorine free radicals?**

(A) Nitrogen dioxide **(B) Methane**

(C) Carbon dioxide **(D) Both (A) and (B)**

Ans. (D)

Nitrogen dioxide and methane act as sink for chlorine free radicals and prevent much ozone depletion.

Q7. **The higher concentration of which gas in air can cause stiffness of flower buds?**

(A) SO_2 **(B)** CO **(C)** NO_2 **(D)** CO_2

Ans. (A)

Due to acid rain, high concentration of SO_2 makes the flower buds stiff and makes them fall off.

Q8. **Ultraviolet light causes**

(A) Formation of pyrimidines

(B) Sticky metaphases

(C) Photodynamic action

(D) Destruction of hydrogen bonds between complementary DNA strands

Ans. (D)

The hydrogen bonds present between complementary strands of DNA are destructed by UV light.

Q9. **White lung disease and black lung disease are caused by**

(A) paper, textiles respectively **(B) textiles, petroleum respectively**

(C) silica, beryllium respectively **(D) asbestos, coal dust respectively**

Ans. (D)

White lung disease is caused by prolonged inhalation of asbestos fibres and black lung disease is caused by coal dust.

Q10 **Green chemistry means such reactions which**

(A) produce chemicals of our daily use from greenhouse gases

(B) study the chemical processes in green plants

(C) perform only those reactions which are related to the depletion of ozone layer

(D) involve use of non-toxic reagents and solvents to produce environmental friendly products.

Ans. (D)

Green chemistry means use of non-toxic reagents and solvents to produce environmental friendly products.

Q11. **The gas that is not considered as a 'green house gas' is**

(A) CO_2 **(B)** CH_4

(C) O_2 **(D)** O_3

Ans. (C)

Oxygen gas does not absorb I.R. radiation of high wavelengths reflected back by earth, hence it does not cause 'green house effect'.

Q12. **Few pollutants and their effects are listed below. Mark the incorrect match.**

(A) Sewage disposed in water - Decrease in DO level

(B) Sulphate fertilizers in water - Eutrophication

(C) Polynuclear aromatic hydrocarbons - Carcinogen

(D) Chlorofluorocarbons - Ozone hole

Ans. (B)

Land run off of nutrients from fertilizers results in the increase of phosphate ions in water which accelerates the formation of algae. Also presence of phosphates accelerates the growth of plants which consume oxygen. As a result, aquatic life is affected and fish is dying. This process is called eutrophication.

Q13. **The living organisms or the materials which are adversely affected by the pollutants is called**

(A) Receptor (B) Sink (C) Target (D) Both (A) and (C)

Ans. (D)

Receptors or targets are organisms or materials which are affected by the pollutants.

Q14. **Which of the following compounds is used in antiknock compositions to prevent the deposition of PbO in different parts of the engine causing damage?**

(A) Acetone **(B) Formaldehyde**

(C) 1, 2-Dibromoethane **(D) Glycerol**

Ans. (C)

1, 2-Dibromoethane is used in antiknock compositions to prevent lead from being deposited in the cylinder head. It is sufficiently volatile and can be removed with the exhaust gases from the exhaust pipe.

Q15. **Which of these are non-biodegradable pollutants?**

(i) DDT **(ii) Mercuric salts**

(iii) Sewage **(iv) Polycyclic aromatic hydrocarbons**

(A) Only (i) and (ii) **(B) Only (i) and (iii)**

(C) Only (i), (ii) and (iv) **(D) Only (iii)**

Ans. (C)

Only sewage is biodegradable pollutant.

Q16 **Which of the following is not correctly matched?**

(A) Fertilizer industries - Gypsum

(B) Thermal power plant - Fly ash

(C) Iron and steel plants - Slag

(D) Rubber industries - Tailings

Ans (D)

Metal industries like Al, Cu, Zn produce mud and tailings.

Q17 **Roasting of sulphide ores give the gas (X) as a by- product. (X) is a colourless gas with choking smell of burnt sulphur and at higher concentration it causes breathlessness and affects larynx. It acts as a reducing agent. Which of the following is correct regarding (X)?**

(A) (X) is SO_2.
(B) Limestone acts as a sink for (X).
(C) (X) can be removed from flue gases by using chemical scrubbers.
(D) All of these.

Ans (D)
All are correct statements.

Q18 **Which compound is mainly responsible for the depletion of ozone layer?**

(A) CO_2 **(B) CH_4**
(C) CH_3OH **(D) CF_2Cl_2**

Ans (D)
Chlorofluorocarbons provide $Cl^{\bullet}$ free radical, which reacts with ozone and converts it into O_2 as.

$$CF_2Cl_2 \longrightarrow Cl^{\bullet} + \dot{C}F_2Cl$$

$$Cl^{\bullet} + O_3 \longrightarrow ClO^{\bullet} + O_2$$

$$ClO^{\bullet} + O \longrightarrow Cl^{\bullet} + O_2$$

The reaction, once start, continues for a long time. Thus, chlorofluorocarbons (CF_2Cl_2) are responsible for the depletion of ozone layer.

Q19 **Loam soil contains almost equal amount of**

(A) silt **(B) Sand**
(C) Clay, humus, air and water **(D)All of these.**

Ans (D)
Loam soil contains almost equal amounts of silt, sand, clay, humus, air and water.

Q20 **Which of the following does not represent a set of air pollutants?**

(A) H_2, N_2, O_2 **(B) SO_2, N_2, CO** **(C) O_3, H_2, CO_2** **(D) NO_x, O_2, O_3**

Ans (A)

H_2, N_2, O_2 are not air pollutants.

Q21. **Harmful chemical present in tobacco is :**

(A) Nicotine **(B) Atropine**

(C) Tannic acid **(D) Morphine**

Ans. (A)

Nicotine is a chiral alkaloid and is used as stimulant and anxiolytic.

Q22. **Which of the following is/are the main agents of soil erosion?**

(A) Wind and water **(B) Rocks** **(C) Sand** **(D) None of thes**

Ans. (A)

Soil erosion is the removal of top layer of soil via wind and water.

Q23. **Depletion of ozone layer is due to**

(A) Oxides of nitrogen **(B) Oxides of carbon**

(C) Oxides of sulphur **(D) None of these**

Ans. (A)

Oxides of nitrogen and chlorofluorocarbons, when reach into the stratosphere, react with ozone molecules and convert them into oxygen. Thus, these compounds lead to depletion of ozone layer.

$$NO + O_3 \longrightarrow NO_2 + O_2$$

$$CF_2Cl_2 \longrightarrow Cl^{\bullet} + \dot{C}F_2Cl$$

$$Cl^{\bullet} + O_3 \longrightarrow ClO^{\bullet} + O_2$$

Q24. **Ozone hole is maximum over**

(A) Europe **(B) Antarctica**

(C) India **(D) Africa**

Ans. (B)

Ozone hole is maximum over Antarctica

Q25. **Depletion of ozone layer over Antarctica takes place**

(A) In November

(B) In the months of September and October

(C) In the months of October and November

(D) In summers

Ans. (B)

During spring season ie, in the month of September and October, the sunlight returs to the Antarctica and breaks up the clouds and photolyses HOCl and Cl_2

$$HOCl \xrightarrow{h\nu} \dot{O}H + \dot{C}l$$

$$Cl_2 \xrightarrow{h\nu} 2\,\dot{C}l$$

These $\dot{C}l$ free radicals again reacts with ozone molecules and leads to ozone depletion

Q26. **As DDT passes into food chain, its concentration**

(A) remains same **(B) decreases**

(C) becomes zero **(D) increases.**

Ans. (D)

With every trophic level in the food chain the concentration of DDT increases.

Q27. **Which one is the most toxic?**

(A) Carbon **(B) CO**

(C) CO_2 **(D) SO_2**

Ans. (B)

Carbon monoxide is highly poisnous to living being.

Q28. **Which one of the following statements is not true?**

(A) Polynuclear aromatic hydrocarbons are organic water pollutants which have carcinogenic effects.

(B) Concentration of DO below 6 ppm is good for the growth of fish.

(C) Clean water would have a BOD value of less than 5 ppm.

(D) Surfactants produce stable foam in rivers.

Ans. (B)

The growth of fish is hindered if concentration of DO is below 6 ppm. Fish dies in water bodies polluted by sewage due to decrease in dissolved oxygen (DO).

Q29. **Monoculture is**

(A) Growing the same crop on the same soil after every two year

(B) Growing different crop on the same soil

(C) Growing the same crop on the same soil after every three year

(D) Growing the same crop on the same soil after one year.

Ans. (D)

Monoculture is the growing the same crop on the same soil after one year.

Q30. **Temperature of troposphere decreases with altitude. This is because of**

(A) High pressure of air **(B) Gases present in air**

(C) Lower density of air **(D) All of these**

Ans. (C)

In troposphere, as we move towards the altitude, the density and pressure of air decreases. Due to which temperature also decreases.

Q31. **Among the following statements which one is incorrect?**

(A) Suspended particulate matter (SPM) is an important pollutant released by diesel vehicles.

(B) Soot particles (size <5 μ) cause fibrosis of the lung lining.

(C) H_2SO_4 particulates have size of 500-1000 nm.

(D) Photochemical smog is formed by oxides of sulphur, smoke and dust par ticles.

Ans. (D)
When climate is warm, dry and sunny, the oxides of nitrogen and unsaturated hydrocarbons are converted in the components such as PAN, formaldehyde which form photochemical smog, by the action of sunlight.

Q32. Black foot disease and minamata disease are caused by respectively
(A) mercury, lead **(B) arsenic, mercury**
(C) lead, arsenic **(D) mercury, arsenic.**

Ans. (B)
Exposure to arsenic via drinking water causes a severe damage to blood vessels, resulting eventually in Foot-gangrene. Minamata disease is neurological disease caused by mercury poisoning.

Q33. Which of the following is responsible for chlorosis?
(A) Cl_2 **(B) NO_2** **(C) SO_2** **(D) O_3**

Ans. (C)
The loss of green colour of plants is called chlorosis. SO_2 is responsible for it.

Q34. Which of the following is secondary pollutant?
(A) CO_2 **(B) N_2O**
(C) PAN **(D) SO_2**

Ans. (C)
Pollutants which are formed by reaction of primary pollutants (persist in the environment in the form they are passed into it) are called as secondary pollutants e.g., peroxyacyl nitrates (PAN) are formed through reaction between nitrogen oxides and hydrocarbons in the presence of sunlight.

Q35. **Water filled in two glasses X and Y have BOD values of 20 and 30 respectively. Select the incorrect statement regarding them.**

(A) X has more DO value than that of Y

(B) Y is more polluted than X

(C) both X and Y are unsuitable for drinking

(D) X is more polluted than Y.

Ans. (D)

Higher the BOD, higher is the pollution.